

Physical Locomotion for Virtual Environments

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ABSTRACT

Physical interaction can provide users with engaging experiences and more healthy active interfaces. In virtual environments, physical interaction can be used for locomotion purposes in which the user's physical actions are mapped to virtual translations, for example in Virtual Reality. Physical engagement can be important not only for exertion game applications, but also in terms of reducing VR sickness, providing engaging and perhaps more realistic experiences, and in the long-term, contribute to reducing sedentary behavior in work context applications. My work examines how locomotion methods can be implemented in virtual environments for various contexts in which the user is physically active in generating locomotion, in contrast to a sedentary desktop or joystick setting. In three studies, I have studied how forms of physical locomotion (normal walking, walk-in-place, fitness equipment) for virtual reality applications, impact the performance and usability within their given context. In these studies I have used comparative experimental design to evaluate locomotion alternatives, or compared with a desktop alternative. For the next study I aim to investigate physical steering techniques for flying experiences. I also aim to synthesize locomotion research into a holistic detail-oriented framework to support comparison and reproducibility between locomotion techniques.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; *HCI theory, concepts and models*.

KEYWORDS

virtual reality, locomotion, physical, embodied, steering

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1 INTRODUCTION

In virtual reality (VR), users navigate through virtually created environments, experiencing the virtual worlds from a first-person perspective. The techniques used to update the first-person viewpoint is termed 'locomotion', and is a key subject of study in VR

research [15, 30, 32, 39]. Over the years there have been numerous ontologies and classification frameworks conceptualizing locomotion along relevant dimensions [15]. One prominent dimension is the degree of "embodiment" or physicality of the locomotion input [15, 30, 37], spanning from low bodily input (e.g. small finger movement) to high involvement of full-body movements. A high degree physical engagement can be important not only for exertion game applications [6, 36, 47], but also in terms of reducing VR sickness [11, 24, 26, 45, 49, 52, 58], providing engaging and perhaps more realistic experiences [4], and in the long-term contribute to reducing sedentary behavior in office work context applications [20].

Aside from the embodied/physical dimension, locomotion techniques have historically been conceptualized along a wide array of dimensions [7, 12, 15, 30, 32, 37, 39]. While these classification provide valid concepts for specific contexts [15], the lack of unified frameworks makes comparisons between locomotion techniques, and reproducing earlier work difficult. Efforts have been made to synthesize all locomotion research into a list of relevant classification variables, e.g. is the locomotion beyond ground, discrete vs continuous, levels of nausea [15]. While these classifications provide valuable high level information about locomotion techniques, they lack in-depth descriptions of *how* locomotion techniques differ.

My initial research studied different physical locomotion methods for specific contexts, including three published studies [21–23], and one more awaiting review. Moving forward (no pun intended), I became curious on how to systematically describe locomotion methods and their associated effects on users. More importantly, I believe the "devil is in the details", and important experiential and performative differences lies in many cases in neglected implementation differences. For example, the amount of acceleration, dampening or rotational speed are important aspects which are typically not described in studies focusing on comparing higher level concepts. Without controlling for, or at least presenting, these detail specifics, study results become harder to trust. In line with efforts to aid research reproducibility [17, 18, 44, 60], an in-depth framework that relates the higher level concepts to implementation level details would help researchers understand and test each others locomotion techniques. This framework should be applicable for locomotion techniques in general, but I will specifically use it as a descriptive tool for my work on physical locomotion methods. This could then serve as a foundation for understanding key implementation details. For example, which details should change when adapting a locomotion technique between more or less physical interaction to increase accessibility.

The main contributions of this research will be:

- (1) A detail-oriented framework for describing all functions involved in converting locomotion input to output.

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- (2) A demonstration by example how the framework can be applied on current and future locomotion implementations.
- (3) An empirical evaluation of the usefulness of the framework for analyzing empirical findings comparing locomotion methods.

2 BACKGROUND AND RELATED WORK

In order to understand how to design for physical locomotion in virtual environments, I have looked at frameworks of locomotion, i.e. what sub-parts does locomotion consists of, what techniques and input modalities can be used, and how previous studies can inform design decisions. Bowman's definition of locomotion in virtual environments broadly consists of three elements: (1) input condition (when and how input is triggered), (2) velocity, and (3) direction [7]. Martinez et al. proposed five categories to consider when designing locomotion in VR [32]: walking-based [34, 46, 48, 50], selection-based (e.g. teleportation), [14, 33, 51], manipulation-based [2, 8, 41], automated travel [29, 54], and steering [15, 32]. Steering addresses direction for locomotion by allowing the user to *continuously* specify the direction of travel. In terms of techniques for input, locomotion in virtual environments is often facilitated by adding external devices and machines to create intuitive ways of locomotion based on real-world experiences such as steering-wheels [25, 27, 43], indoor bicycles [5, 9, 35], or leaning-based seats [1, 57]. These external devices can also provide more physical interaction experience compared with joystick-based locomotion for example. In two of my studies (see 1) I have examined locomotion techniques where the user is restricted from walking freely in the virtual environment, i.e. the velocity and/or direction is not set by a direct mapping of the user's physical movement to the virtual. This requires some form of *steering technique* to be implemented. Several steering techniques have been proposed through the years [32], mainly for ground-based 2D locomotion [31]. These techniques are typically categorized based on which body part controls input and created five groups based on previous literature: *head*, *torso*, *hands*, *legs*, and *feet*. In "torso-directed" steering, the user would set the travel direction based on the torso's orientation [42, 59]. Leg motion for steering has, for example, been implemented for flying [1, 57] and scuba diving [28] in VR.

3 RESEARCH TO DATE

A first short paper titled "Creating a Post-Sedentary Work Context for Software Engineering" [20] outlines the initial plan of studying various aspects of software engineering in a post-sedentary context. In the beginning I wanted to focus on the experiential, motivational aspects of physical interaction in a work context. Since my intended PhD field was (and is) within interaction programming I tried to marry software engineering activities to 3D-interaction and movement. In other words, how can software engineering activities be carried out in using physical interfaces instead of a sedentary desktop setting. For multiple reasons the emphasis became fully on the health aspects of reducing sedentary behavior. The papers describes different technologies that can be used to achieve this purpose, with Virtual/Mixed Reality being one of the primary solutions.

The second and third paper was published at MUM 2023 [22, 23]. The second paper is about post-sedentary block-based programming, and the third paper is a comparative study on walk-in-place methods adapted from my master thesis. The block-based paper builds upon the post-sedentary concept outlined in the first paper by implementing a block-based programming interface for virtual reality. A user study was conducted which compared the VR interface to a desktop version in terms of efficiency. Participants physical activity was also measured in the VR version, which provided a descriptive picture on how *much*, and during *which* user activities movement occurred. In hindsight, as some reviewers pointed out, it would have been interesting to explore more experiential qualities of the interface in the study. The third paper compares 3 walk-in-place methods for virtual reality where the user's have to "jog-in-place" to travel in the virtual environment. The methods differ in how steering is done, using a head-, hand- or torso-based approach to determine the direction of travel. The study was conducted before my PhD started but is topically related through the shared focus on physically active interaction and virtual reality. My forth paper [21] just got accepted to CHI'24 and it explores locomotion using rowing machines, and a user study for a fifth paper exploring physical locomotion for flying in VR is currently scheduled.

4 APPROACH

For the two upcoming papers (one under review, one at user study phase) I have compared physical locomotion techniques based on different input modalities. The first paper uses a gym fitness device as input source while the other uses a leaning-based [15] approach while only relying on VR controllers. In the next paper I aim to reflect on my previous implementations and develop a framework for locomotion techniques. While introducing new concepts and classification could contribute to the "toothbrush" problem [16] (i.e. nobody wants to use someone else's toothbrush), I believe my approach to framework construction is complementary to established frameworks as its focus will be on implementation details rather than new high level abstractions. I have begun sketching this framework which can be seen in figure 2 and will use this section to describe it.

The proposed framework conceptualizes locomotion as a set of functions converting an input to *position* and *orientation* changes of the first-person perspective. The seven steps in this process are categorized as 1) *source*, 2) *processing*, 3) *classification*, 4) *external forces*, 5) *previous input*, 6) *mapping*, 7) *output*. Source describes what original data type the locomotion input consists of, for example a discrete boolean value, a continuous range or vectors. It also describes what hardware was used to record the user's physical motion into a digital representation (e.g. a specific IMU sensor or a specific motion capture such as Microsoft Kinect), where the hardware is located (e.g. how the user holds it or location in a room), and how the user interacts with the hardware (e.g. pressing with thumb, reposition with one hand). In the next step (processing), sampling techniques (e.g. Kalman's filter) can be used to filter out noise and thresholds can be used to limit the input to reasonable ranges. The classification step acts as a route and gatekeeper, channeling processed input to targeted output. For example, ML algorithms

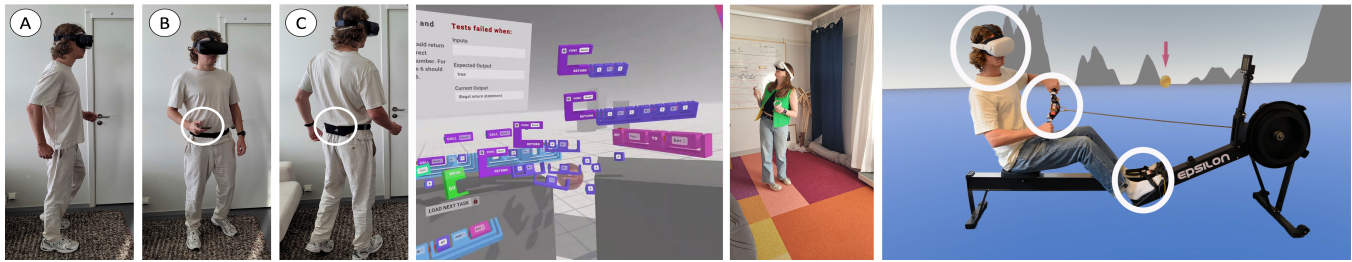


Figure 1: Previous studies of physical locomotion techniques: walk-in-place, walking, rowing

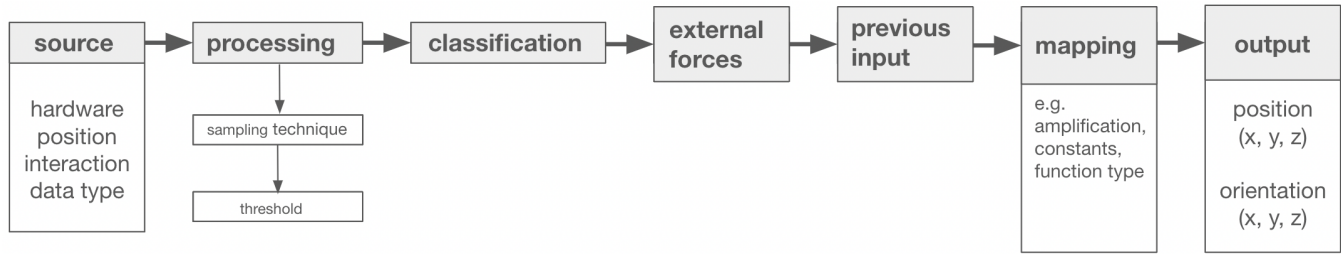


Figure 2: Current iteration of proposed locomotion technique framework. The whole framework can be described as a function which modifies incoming input at each step and outputs the new position and orientation of the first-person viewpoint.

classifies inputs into motion patterns which should trigger a specific locomotion behavior, or as in my “jogging-in-place” technique [23] an algorithm determines if inputs should be classified as “jogging steps” or not.

When an output source have been targeted, the next three steps modifies the processed input. The first of these steps alters the input depending on external forces such as gravity, winds, collisions or surface material. The second step first determines if previous locomotion should be used for computing current output. If true, this step describes specific algorithms for output altering algorithms such as acceleration and dampening. Lastly, the mapping step computes the final output for a desired target (e.g. rotate the user’s view around “yaw” axis), through specific algorithms (e.g. power functions, constant and scalar values). By describing each step, a more complete and detail-oriented picture of locomotion techniques can be established, supporting comparisons, new discoveries and reproducing other researchers techniques. While sharing source code is also important, it can often be complex, time-consuming and error-prone to understand and run other’s projects. A detail-oriented form would instead allow researchers to implement others locomotion techniques in a tutorial like manner. In the following time-period, my aim is to finalize this framework, demonstrate its applicability on my four physical locomotion methods, and then study if others can replicate locomotion methods by coding and decoding their locomotion techniques using the framework.

4.1 Alternative routes

Besides developing, demonstrating and evaluating the locomotion framework, I consider two other approaches to continue my research within physical locomotion methods. One approach focuses

on the impact of walking or body repositioning in work contexts, while the other takes a perceptual psychological approach.

4.1.1 Physical Activity in Work Context. This route builds upon my initial interests and could fit within an overall thesis theme. The core idea is to include physical activity while otherwise sedentary activities would take place. A good example paper would be the “walking-stick” [19], a prototype for facilitating meetings while walking. I have previously explored various mid-air text-entry method techniques, and I believe this is a key path forward, for example by combining with augmented reality glasses. An example study would be to evaluate *how* productivity is impacted for text-entry tasks (e.g. answering emails) while walking outdoors using different text-input methods with AR-glasses.

4.1.2 Anxiety and Movement. This research interest is related to my initial interest in exploring the role of physical engagement for experience and meaningfulness in work context interaction. The idea stems from my observations and experiences as a software developer: most of my colleagues are/or have become bland from working. I also noticed that I performed better if my energy levels were lower while programming, allowing me to concentrate and stay focused. I have also noticed the common habit of using some form of nicotine as focus enhancing substance while working in the software industry. I noticed how both feelings of frustration and engagement while programming caused anxiety, which for me could be better tolerated with nicotine. Recent research suggests that feeling anxious makes you want to physically move and that anxiety is only experienced as uncomfortable when forced to remain still [55]. My hypothesis is that frustration or engagement stemming from goal-directed intentions will signal to you through emotions that you should move. Canceling this urge is cognitively

demanding. I would like to investigate *if* this hypothesis is true, and *if* alternative physically demanding interfaces that requires more physical effort decreases feelings of anxiety, and *how* performance thereby is impacted.

5 QUESTIONS

Even though I have finished three years of my PhD, I am very open for new directional changes and input. The theme of physical locomotion functions as an umbrella term of my previous studies and the developed framework felt as a natural next step to synthesize the studies into a meaningful whole. I am therefore curious on methods and approaches to validate frameworks. I am also open to the idea of moving in the opposite direction, focusing on one specific locomotion technique and conducting new user studies. In such case I would appreciate ideas and feedback on application scenarios and interesting angles, perhaps bridging the locomotion research into other realms of HCI inquiry.

6 DISSERTATION STATUS AND LONG-TERM GOALS

My PhD program is five years in total, including one full year of courses and one full year of teaching. I have just completed my third year, and have completed all teaching duties (including hosting and teaching the course "Advanced Interaction Programming", which I developed myself for final year Master students), and I have completed half of my courses. I have finished my 50% seminar and thus would consider myself about 50% complete, depending on which publications (four in total) I include in the dissertation. My long-term goal for an academic career is uncertain, as I am grappling with concerns of academic incentive structures [3, 38, 53], reproducibility [17, 56, 60] and translational aspects [10, 13, 40]. I aim to finish during fall 2025.

7 BENEFIT AND CONTRIBUTIONS

I would greatly benefit from attending the doctoral consortium at CHI 2024 as my research focus has shifted over the years. Therefore, receiving feedback, suggestions and inspiration for how to best conceptualize my work is valuable for me. Additionally, I am very interested in "meta scientific" research questions, and would be curious to understand the experiences of other doctoral students in the HCI community. Lastly, the benefit of networking with highly motivated and competent people is a "no-brainer". I am an optimistic, fun and highly driven doctoral student and will contribute through my unique experiences and research approach. I am not afraid to speak my mind but will do so in a respectful manner. Most importantly, I think other doctoral students would benefit from understanding the mistakes I've made and why they happened.

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